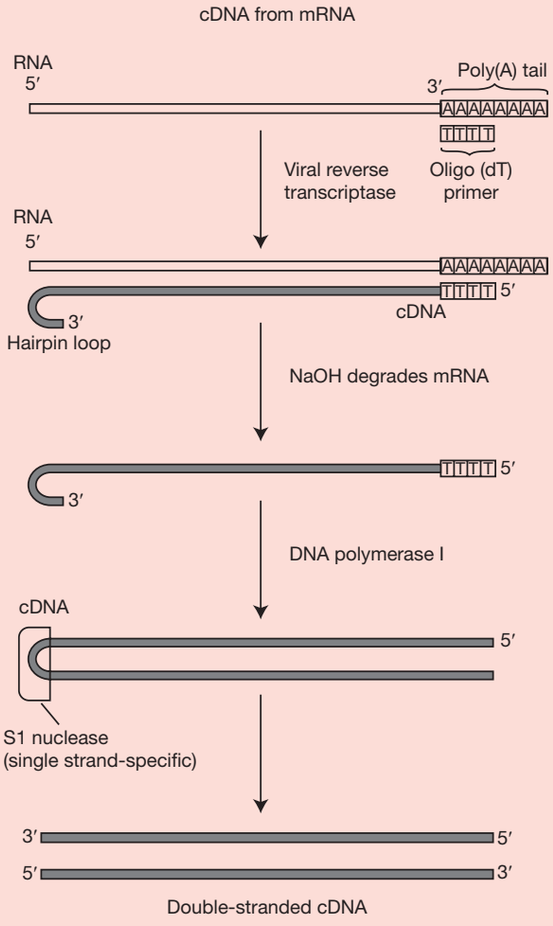
Infobits

In 1970 Howard Temin and David Baltimore independently discovered the enzyme **reverse transcriptase** that retroviruses use to produce DNA copies of their RNA genome. This enzyme can be used to construct DNA copy, called **complementary DNA** (cDNA), of any RNA as shown in figure below . Thus genes or major portions of the gene can be synthesized from mRNA.



of an organism. **Phenotype** is an observable property of organism. The functional form of a gene is called **Wildtype** because presumably this is the form found in nature.

Mutation is the process by which the sequence of base pairs in a DNA molecule is altered.The alteration can be a single base pair substitution, insertion or deletion.

Mutations can be divided into two general categories:

1. **Base–pair substitution mutation** involves a change in the DNA such that one base pair is replaced by another.
   - A mutation from one purine – pyrimidine base pair to the other purine –pyrimidine base pair is a transition mutation (Figure 12.7a). E.g. AT to GC, CG to TA.
   - A mutation from a purine - pyrimdine base pair to a pyrimidine – purine base pair is a transversion mutation (Figure 12.7b). E.g. AT to TA, CG to GC.
2. **Base pair insertion or deletions** involves the addition or deletion of one base pair. If one or more base pairs are added to or deleted from a protein coding gene, the reading frame of an mRNA can change downstream of the mutation. An addition or deletion of one base pair, for example, shifts the mRNA's downstream reading frame by one base, so that incorrect amino acids are added to the polypeptide chain after the mutation site. This type of mutation, called a frame shift mutation (Figure 12.8) usually results in a nonfunctional protein.

Frame shift mutations:

- May generate new stop codons, resulting in a shortened protein.
- May result in a read through of the normal stop codon, resulting in longer than normal proteins
- Or may result in a complete alteration of the amino acid sequence of a protein.

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